

VARRSHA M

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SUMMARY

Aspiring student of data science and machine learning with practical experience from personal and academic projects in predictive analytics, computer vision, and healthcare AI. competent in preprocessing data, developing machine learning models, assessing outcomes, and developing straightforward deployment APIs or interfaces. eager to learn how to transform real-world data into insightful conclusions and workable AI solutions.

PROJECTS

CARDIO DETECT (KBIS + ML)

[Source code](#)

- Developed a hybrid ML system integrating U-Net segmentation, ensemble learning, and KBI scoring for ~70% accurate blockage detection.
- Processed real hospital datasets, engineered clinical features, and validated model stability using cross-validation.
- Integrated a Streamlit interface for real-time prediction & explainable diagnostics.

GENE-DISEASE CLASSIFICATION & DRUG REPURPOSING

[Source code](#)

- Built ML pipelines (XGBoost, RF) for multi-class gene expression classification (~90% accuracy).
- Identified co-expressed genes using TFF, IFNG pathway clustering & patient segment analysis.
- Used curated datasets from DrugBank, KEGG & Government repositories for biomarker exploration.

SMARTCAMPUS ID DETECTOR

[Source code](#)

- Designed & deployed a CNN-ANN hybrid model for multi-class ID detection with 70:30 stratified split, achieving ~99% accuracy.
- Implemented robust preprocessing, class balancing, and glare/lighting correction for real-world data.
- Built a modular ML framework enabling seamless scaling with YOLO-based extensions.

SAFEZONE AI

[Source code](#)

- Developed SafeZone AI, an IoT-ML system using Random Forest and Arduino sensors to detect heat, fatigue, proximity, and access hazards in real time.
- Integrated with AWS Iot Core and DynamoDB for cloud alerts streamlist dashboard for real-time monitoring

EDUCATION

Integrated Masters in Artificial Intelligence & Machine Learning

Coimbatore Institute of Technology , Coimbatore

Expected Graduation, July 2028

CGPA 8.72/10 as of 4th sem

Higher Secondary (HSE+2)

Chandra Matric Higher Secondary School , Coimbatore

March 2023

Percentage : 92%

SKILLS

Languages

Python, MYSQL,C, HTML, CSS

ML / AI Tools & Libraries

Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn

AI & Machine Learning Concepts

Advanced Machine Learning, Artificial Neural Networks , Data Structures

Data Engineering & Deployment

Flask, Streamlit, Rest API design, AWS IoT Core, DynamoDB, Git

Soft Skills

Leadership,Communication, Time management, Resilience

INTERESTS

- Data Science
- Machine Learning
- Predictive Analytics
- Statistical Modeling

CERTIFICATIONS

- [Business Intelligence & Analytics \(NPTEL\)](#) – 12-week course on data analysis and visualization.

Jan-Apr 2025

[Link](#)

EXPERIENCE & ACHIEVEMENTS

- Earned Special Mention at [KLA UNHACK](#), an inter-college, milestone-based hackathon organized by KLA Corporation.
- Received Special Mention in [CODE VORTEX](#), conducted by the Department of Computing, CIT, in collaboration with KLA Alumni.
- [EVENT ORGANIZING](#) : Co-organized CISKa Marathon for the past 3 years (2000+ participants)
- [ATHLETICS](#) : Zonal Gold & District Bronze medalist

[Link](#)

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